

A cheetah pursues a Thomson’s gazelle. The predator–prey equations are able to model the way populations of both species will change in response to the activities of the other.

“The food species cannot, therefore, be exterminated by the predatory species, under the conditions to which our equations refer.”

Alfred J. Lotka

Population principles

At the time Lotka and Volterra made their calculations, the science of population dynamics was still in its infancy, having barely moved on since the population studies of British economist Thomas Malthus in the late 18th century. According to Malthus’s theory, a population grows or declines rapidly as long as the environmental factors for survival are constant, and the rate at which that population changes increases as the population grows. From this theory, Malthus predicted a catastrophic future for humanity. The number of humans was growing much more quickly than the amount of food that could be produced by the world’s farmlands. Eventually, Malthus argued, a point would be reached when the human population would succumb to global famine and decline.